Chinmayi K

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.58
NARAYANA OLYMPIAD SCH SAHAKARANAGAR BANGALORE KK	Class XII (CBSE)	2024	90.20%
VISHWACHETANA V NIKETANA RES ENG M SCH DAVANGERE	Class X (CBSE)	2022	97.00%

SKILLS

Languages: Python, SQL, C, C++

Databases: SQLite, MySQL

Development Tools: Jupyter Notebook, Figma, Canva, Git/Github, Visual Studio Code, MATLAB, CMake

Frameworks/Libraries: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Statsmodels, XGBoost, Imbalanced-learn

PROJECTS

Thread-Safe LRU Cache:

C++, Multithreading, CMake, Object-Oriented Programming (OOP)

- Built an O(1) LRU Cache in C++ using a hash map and a doubly linked list, supporting efficient cache lookup, updates and least recently used eviction.
- Implemented thread-safe concurrent access using C++ synchronization primitives to prevent race conditions during simultaneous cache operations.
- Developed multithreaded stress tests and performance benchmarks to evaluate throughput and cache hit/miss ratios under varying cache capacities and concurrency levels.

Credit Card Fraud Detection & Risk Analytics:

Python, SQL, SQLite, Pandas, Scikit-learn, XGBoost, Imbalanced-learn (SMOTE), Matplotlib

- Queried and analyzed a 284K+ transaction dataset using SQL and Python to investigate fraud patterns and severe class imbalance (0.17% fraud prevalence).
- Compared class weighting, SMOTE oversampling and random undersampling to address class imbalance, evaluating model performance using Precision-Recall AUC instead of accuracy.
- Built Logistic Regression and XGBoost models, analyzed feature importance and selected a classification threshold based on fraud detection performance and business trade-offs.

Product Feature Rollout A/B Test Analysis:

Python, SQL, SQLite, Pandas, NumPy, SciPy, Statsmodels, Matplotlib

- Queried and aggregated data for a 90K+ user A/B experiment using SQL before analyzing the impact of a gameplay feature rollout on user retention and engagement using Python.
- Performed Sample Ratio Mismatch (SRM) validation, power analysis and hypothesis testing using chi-square and Mann-Whitney U tests to compare experiment groups.
- Measured statistical and practical significance using effect size estimation and conducted engagement-based segmentation to support data-driven product recommendations.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Algorithms

Mathematics: Statistics, Probability, Calculus, Differential Equations, Elementary Linear Algebra, Complex Variables

POSITIONS OF RESPONSIBILITY

Design Head, Echo (Media Council, IIT Hyderabad) June 2026 - Present

- Design official IIT Hyderabad merchandise and create branding assets and promotional content for the Student Gymkhana, ensuring consistent and engaging visual communication.
- Lead the design team in developing creatives for institute-wide events, coordinating with different student bodies and managing end-to-end design execution using Figma.

Outreach and Collaborations Coordinator, TEDxIITHyderabad June 2025 - May 2026

- Built partnerships with schools, colleges and organizations to expand the reach of TEDxIITHyderabad.
- Worked closely with the organizing team to manage outreach and contribute to the successful execution of TEDx events.

EXTRACURRICULAR

Inter-IIT Cultural Meet 8.0 | IIT Kanpur

- Represented IIT Hyderabad at the **Inter-IIT Cultural Meet 8.0** in Kanpur as a member of the **Group Dance** contingent.
- Competed in the **Duet Dance** event at the Inter-IIT Cultural Meet 8.0.

Inter-IIT Cultural Meet 7.0 | IIT Patna

- Represented IIT Hyderabad in the **Group Dance** event at the **Inter-IIT Cultural Meet 7.0** held at IIT Patna.

Dance Achievements

- Secured **3rd Place** in the **Solo Dance** competition at **Elan & nVision**, the annual techno-cultural fest of IIT Hyderabad (2026).
- Won **1st Place** in the **Group Dance** competition at **Diesta**, the inter-departmental fest of IIT Hyderabad (2025).
- Won **1st Place** in the **Trio Dance** competition at **Diesta**, the inter-departmental fest of IIT Hyderabad (2025).